

people suffering from P-type and D-type colour blindness to distinguish between red and green, so that correct information is accurately interpreted by them.

ROHM Semiconductor
www.rohm.com

Secure Bluetooth solutions

These Bluetooth Low Energy SoCs and modules are ultra-compact, low-cost and have long battery life as well as security

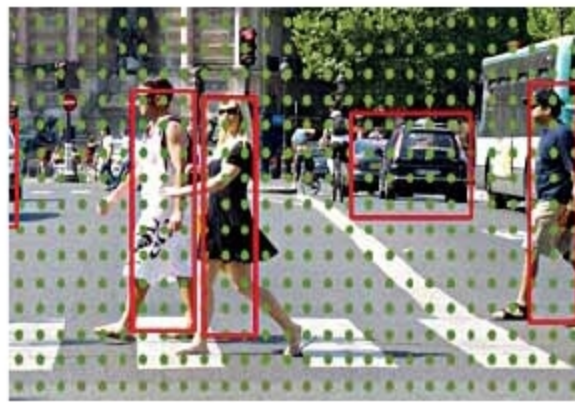


features that include Secure Boot with Root of Trust and Secure Loader, which, along with high-performance and security, are ideal for IoT device makers. Due to small size, the SiP modules add turnkey Bluetooth connectivity to extremely small products. These also support multiprotocol connectivity for the most demanding applications ranging from gateways, hubs, and smart lighting to wireless sensors, actuators, and portable medical equipment. Their small size and fully certified bluetooth SiP allows designing devices with Bluetooth capability without worrying about space generally scarce in wearables and earphones. With a small SiP, more free space is available to accommodate batteries and other sensors.

Silicon Labs
www.silabs.com

4D imaging radar on a chip

This all-weather radar can detect cars from 500 metres and pedestrians from 200 metres, with an angular resolution greater than one degree. It uses a powerfully focused beam based on proprietary Phased Array radar technology, which is used in cutting-edge military systems including the F-35 fighter jet and air-defence systems. The focused beam created by dozens of transmitters rapidly



scans the field of view. The receivers ensure a much-improved radar image, a better signal-to-noise ratio, and a wide detection range (nearly 6x compared to existing radars) of obstacles such as cars and pedestrians. Using advanced military-grade technology used in commercial automotive applications, it ensures robust and reliable safety of both pedestrians and passengers.

RFISee
www.rfisee.com

TMR digital magnetic sensor

RedRock RR122-1A22-511/512 wafer-based TMR (Tunneling Magnetoresistance) digital omnipolar sensor has operating frequency of 10Hz, a voltage



supply range of +1.7V to +5.5V. With a low current drain (75nA) for an active magnetic sensor, the RR122-1A22 series is suited for performing the 'wake up' or 'turn on' function for battery operation. It is optimal for power-sensitive applications, and offers extremely strong magnetic sensitivity (9G or 0.9mT).

Coto Technology
www.cotorelay.com

Thin optical sensor

The MAXM86146, a dual-photodetector optical sensor, includes built-in algorithms for heart-rate and oxygen-saturation (SpO2) monitoring, as well as advanced activity classification. Its small size and ease of design allows innovative biosensing features in less space and supports rapid development

of accurate, continuous health-monitoring devices. Additionally, the module comes with biosensing algorithms that



are verified by strict medical standards for measuring vital signs. It allows wearable product designers to achieve a 45% thinner optical design and provides ease of access for health monitoring.

Maxim Integrated
www.maximintegrated.com

Chip that requires no battery

An ultra-low-power MCU called the ONiO.zero, which can harvest energy from its surroundings and utilise it, requires no external batteries or capacitors. Despite the small size, it is quite powerful and can operate with the help



of a wide range of energy sources such as multi-frequency RF bands including GSM and ISM. Built on the RISC-V 16/32 bit architecture (RV32EMC), it has a mask ROM of 1kB, RAM of 2kB, and ULP Flash of 8/16/32kB. Operating frequency is up to 24MHz, while it can safely work under temperatures ranging from -40 to 85 degrees Celsius. An internal RF rectifier of multi-frequency bands (800/900/1800/1900/2400 MHz) provides support for harnessing power from GSM or ISM. It can achieve high energy efficiency of 85 to 90%. No batteries means less number of components required, which leads to more space.

ONiO
www.onio.com